

OS-OP

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1.

```
sadwika@DESKTOP-8CQA01L X + ~
GNU nano 8.7.1 week1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];
    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

```
sadwika@DESKTOP-8CQA01L:~$ nano week1.c
sadwika@DESKTOP-8CQA01L:~$ gcc week1.c -o week1
sadwika@DESKTOP-8CQA01L:~$ ./week1
Enter Linux command: pwd
Parent PID : 997
Child PID : 998
/home/sadwika
Child process completed.
sadwika@DESKTOP-8CQA01L:~$ gcc week1.c -o week1
sadwika@DESKTOP-8CQA01L:~$ ./week1
Enter Linux command: ls
Parent PID : 1022
Child PID : 1023
S2Commands      exec      hello.c      username     wait1
a.out            exec.c     parentchild  username.c   wait1.c
displayPID       fork      parentchild.c wait          week1
displayPID.c     fork.c    process     wait.c       week1.c
example.txt      hello     process.c   wait.c.save
Child process completed.
```

```
sadwika@DESKTOP-8CQA01L:~$ uname -a
Linux DESKTOP-8CQA01L 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
sadwika@DESKTOP-8CQA01L:~$ uname -r
6.18.33.2-microsoft-standard-WSL2
sadwika@DESKTOP-8CQA01L:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       12
On-line CPU(s) list:         0-11
Vendor ID:                    GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i7-1355U
CPU family:                   6
Model:                        186
Thread(s) per core:           2
Core(s) per socket:           6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     5222.39
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic s
                             ep mtrr pge mca cmov pat pse36 clflush mm
                             x fxsr sse sse2 ss ht syscall nx pdpe1gb
                             rdtscp lm constant_tsc rep_good nopl xtop
                             ology tsc_reliable nonstop_tsc cpuid tsc_
                             known_freq pni pclmulqdq vmx ssse3 fma cx
                             16 pcid sse4_1 sse4_2 x2apic movbe popcnt
                             tsc_deadline_timer aes xsave avx f16c rd
                             rand hypervisor lahf_lm abm 3dnowprefetch
                             ssbd ibrs ibpb stibp ibrs_enhanced tpr_s
                             hadow ept vpid ept_ad fsgsbase tsc_adjust
                             bml avx2 smep bmi2 erms invpcid rdseed
                             adx smap clflushopt clwb sha_ni xsaveopt
                             xsavec xgetbv1 xsaves avx_vnni vnni umip
                             waitpkg gfni vaes vpclmulqdq rdpid movdir
                             i movdir64b fsrm md_clear serialize ibt f
                             lush_lld arch_capabilities

Virtualization features:
```

```
Virtualization features:
  Virtualization:             VT-x
  Hypervisor vendor:           Microsoft
  Virtualization type:         full
Caches (sum of all):
  L1d:                         288 KiB (6 instances)
  L1i:                         192 KiB (6 instances)
  L2:                           7.5 MiB (6 instances)
  L3:                           12 MiB (1 instance)
NUMA:
  NUMA node(s):                1
  NUMA node0 CPU(s):           0-11
Vulnerabilities:
  Gather data sampling:         Not affected
  Ghostwrite:                   Not affected
  Indirect target selection:     Not affected
  Itlb multihit:                Not affected
  Lltf:                         Not affected
  Mds:                          Not affected
  Meltdown:                     Not affected
  Mmio stale data:              Not affected
  Old microcode:                Not affected
  Reg file data sampling:        Mitigation; Clear Register File
  Retbleed:                     Mitigation; Enhanced IBRS
  Spec rstack overflow:          Not affected
  Spec store bypass:            Mitigation; Speculative Store Bypass disa
                                bled via prctl
  Spectre v1:                   Mitigation; usercopy/swapgs barriers and
                                __user pointer sanitization
  Spectre v2:                   Mitigation; Enhanced / Automatic IBRS; IB
                                PB conditional; PBRSE-eIBRS SW sequence;
                                BHI BHI_DIS_S
  Srbds:                        Not affected
  Tsa:                          Not affected
  Tsx async abort:              Not affected
  Vmscape:                      Not affected
sadwika@DESKTOP-8CQA01L:~$ lscpu | head -20
Architecture:                x86_64
```

```
sadwika@DESKTOP-8CQA01L:~$ lscpu | head -20
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-1355U
CPU family:             6
Model:                 186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:               5222.39
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss h
syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx
6 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp
ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt x
avec xgetbv1 xsave avx_vnni vnni unpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_clear serialize ibt flush_l1d arch_capabilities
ies
Virtualization:         VT-x
Hypervisor vendor:      Microsoft
Virtualization type:    full
L1d cache:              288 KiB (6 instances)

sadwika@DESKTOP-8CQA01L:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    disk [SWAP]
sdd   8:48   0    1T    disk /mnt/wslg/distro

sadwika@DESKTOP-8CQA01L:~$ lsblk -f
NAME FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda ext4    1.0
sdb ext4    1.0
sdc swap    1          8500dc33-c992-46d6-8842-7e6992c88cfc [SWAP]
```

```
sadwika@DESKTOP-8CQA01L:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1   24420 15428 ?        Ss   04:05   0:03 /sbin/init
root         2  0.0  0.0   3180  2208 hvc0    Sl+  04:05   0:00 /init
root         6  0.0  0.0   3180  2044 hvc0    Sl+  04:05   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-trunc
root        52  0.0  0.2  50396 16936 ?        S<s  04:05   0:00 /usr/lib/systemd/systemd-journald
systemd+   84  0.0  0.1  22540 14604 ?        Ss   04:05   0:00 /usr/lib/systemd/systemd-resolved
root       100  0.0  0.1  35360 12280 ?        Ss   04:05   0:02 /usr/lib/systemd/systemd-udevd
root       156  0.0  0.0   2888  1952 ?        Ss   04:05   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root       157  0.0  0.0   4472  3144 ?        Ss   04:05   0:00 /usr/sbin/cron -f -P
message+   158  0.0  0.0   8788  5368 ?        Ss   04:05   0:00 dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activ
root       166  0.0  0.3  44092 29856 ?        Ss   04:05   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root       178  0.0  0.1  18432  9408 ?        Ss   04:05   0:00 /usr/lib/systemd/systemd-logind
syslog     209  0.0  0.0  220548 5928 ?        Ssl  04:05   0:00 /usr/sbin/rsyslogd -n -iNONE
root       278  0.0  0.0   5492  2756 tty1    Ss+  04:05   0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/ru
_chrony    332  0.0  0.1  24116 11656 ?        S   04:05   0:00 /usr/sbin/chronyd -n -F 1 -x
root       343  0.0  0.4 123456 32684 ?        Ssl  04:05   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown -
_chrony    344  0.0  0.0 12092  2452 ?        S   04:05   0:00 /usr/sbin/chronyd -n -F 1 -x
root       417  0.0  0.0   8644  5572 ?        Ss   04:05   0:00 login -- sadwika
root       420  0.0  0.0   3188  1124 ?        Ss   04:05   0:00 /init
root       421  0.0  0.0   3204  1260 ?        S   04:05   0:00 /init
sadwika    422  0.0  0.0   6268  5540 pts/2    Ss   04:05   0:00 -bash
sadwika    471  0.0  0.1  22336 13512 ?        Ss   04:05   0:00 /usr/lib/systemd/systemd --user
sadwika    473  0.0  0.0  22912  4080 ?        S   04:05   0:00 (sd-pam)
sadwika    502  0.0  0.0   6136  5416 pts/1    Ss+  04:05   0:00 -bash
root      1287  0.0  0.1 1792044 15364 ?        Ssl  05:08   0:00 /usr/libexec/wsl-pro-service
root      1632  0.0  0.2 340136 22392 ?        Ssl  05:10   0:00 /usr/libexec/packagekitd
polkitd   1637  0.0  0.1 306444  8464 ?        Ssl  05:10   0:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
sadwika   1715  0.0  0.0   7160  4364 pts/2    R+   05:12   0:00 ps aux

sadwika@DESKTOP-8CQA01L:~$ top
top - 05:13:35 up 1:07, 1 user, load average: 0.00, 0.04, 0.00
Tasks: 27 total, 1 running, 26 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.6 total, 7143.2 free, 556.5 used, 239.0 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7228.2 avail Mem
```

```
PID USER PR NI VIRT RES SHR S %CPU %MEM TIME+ COMMAND
1725 sadwika 20 0 8168 5996 3900 R 0.3 0.1 0:00.04 top
1 root 20 0 24420 15428 11604 S 0.0 0.2 0:03.22 systemd
2 root 20 0 3180 2208 2072 S 0.0 0.0 0:00.00 init-systemd(Ub
6 root 20 0 3180 2044 1940 S 0.0 0.0 0:00.00 init
52 root 19 -1 50396 16940 15620 S 0.0 0.2 0:00.57 systemd-journal
84 systemd+ 20 0 22540 14604 11984 S 0.0 0.2 0:00.17 systemd-resolve
109 root 20 0 35360 12280 9192 S 0.0 0.2 0:02.36 systemd-udev
156 root 20 0 2888 1952 1820 S 0.0 0.0 0:00.13 chronyd-starter
157 root 20 0 4472 3144 2892 S 0.0 0.0 0:00.06 cron
158 message+ 20 0 8788 5368 4672 S 0.0 0.1 0:00.22 dbus-daemon
166 root 20 0 44092 29856 14716 S 0.0 0.4 0:00.67 networkd-dispat
178 root 20 0 18432 9408 8160 S 0.0 0.1 0:00.23 systemd-logind
209 syslog 20 0 220548 5928 4632 S 0.0 0.1 0:00.16 rsyslogd
278 root 20 0 5492 2756 2572 S 0.0 0.0 0:00.02agetty
332 _chrony 20 0 24116 11656 7124 S 0.0 0.1 0:00.24 chronyd
343 root 20 0 123456 32684 18076 S 0.0 0.4 0:00.33 unattended-upgr
344 _chrony 20 0 12092 2452 1808 S 0.0 0.0 0:00.01 chronyd
417 root 20 0 8644 5572 4732 S 0.0 0.1 0:00.05 login
420 root 20 0 3188 1124 980 S 0.0 0.0 0:00.00 SessionLeader
421 root 20 0 3204 1260 1108 S 0.0 0.0 0:00.28 Relay(422)
422 sadwika 20 0 6268 5540 3864 S 0.0 0.1 0:00.40 bash
471 sadwika 20 0 22336 13512 10900 S 0.0 0.2 0:00.35 systemd
473 sadwika 20 0 22912 4080 2076 S 0.0 0.1 0:00.00 (sd-pam)
502 sadwika 20 0 6136 5416 3864 S 0.0 0.1 0:00.11 bash
1287 root 20 0 1792044 15364 12560 S 0.0 0.2 0:00.09 wsl-pro-service
1632 root 20 0 340136 22392 19336 S 0.0 0.3 0:00.07 packagekitd
1637 polkitd 20 0 306444 8464 7652 S 0.0 0.1 0:00.10 polkitd

sadwika@DESKTOP-8CQA01L:~$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
sadwika@DESKTOP-8CQA01L:~$ lspci
Command 'lspci' not found, but can be installed with:
```

```
sadwika@DESKTOP-8CQA01L:~$ lspci
1fbd:00:00.0 3D controller: Microsoft Corporation Basic Render Driver
5582:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio 1.0 console (rev 01)
ee16:00:00.0 System peripheral: Red Hat, Inc. Virtio 1.0 file system (rev 01)
```

2.

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```
sadwika@DESKTOP-8CQA01L:~$ nano week.c
sadwika@DESKTOP-8CQA01L:~$ gcc week.c -o week
sadwika@DESKTOP-8CQA01L:~$ ./week
Cannot open source file
sadwika@DESKTOP-8CQA01L:~$ nano input.txt
sadwika@DESKTOP-8CQA01L:~$ gcc input.txt -o input.txt
gcc: fatal error: input file 'input.txt' is the same as output file
compilation terminated.
sadwika@DESKTOP-8CQA01L:~$ ./week.c
-bash: ./week.c: Permission denied
sadwika@DESKTOP-8CQA01L:~$ ./week
File copied successfully.
sadwika@DESKTOP-8CQA01L:~$ cat output.txt
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
```

```
int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);

    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
```

```
    }

    close(src);
    close(dest);

    printf("File copied successfully.\n");

    return 0;
}
```

```

sadwika@DESKTOP-8CQA01L:~$ strace cat input.txt
execve("/usr/bin/cat", ["cat", "input.txt"], 0x7ffed188a7d8 /* 25 vars
*/) = 0
brk(NULL) = 0x5b30ee7a8000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1,
0) = 0x7ebc4ffb1000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or d
irectory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20151, ...}) = 0
mmap(NULL, 20151, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7ebc4ffac000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY
|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\
0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7eb
c4ff79000
mmap(0x7ebc4ff8000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXE
D|MAP_DENYWRITE, 3, 0x7000) = 0x7ebc4ff80000
mmap(0x7ebc4ffa1000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYW
RITE, 3, 0x28000) = 0x7ebc4ffa1000
mmap(0x7ebc4ffa8000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
|MAP_DENYWRITE, 3, 0x2f000) = 0x7ebc4ffa8000
mmap(0x7ebc4ffaa000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
|MAP_ANONYMOUS, -1, 0) = 0x7ebc4ffaa000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O
_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\
0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7eb
c4ff4b000
mmap(0x7ebc4ff4f000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXE
D|MAP_DENYWRITE, 3, 0x4000) = 0x7ebc4ff4f000

```

```

close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7e
bc4fe25000
mmap(0x7ebc4fe37000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXE
D|MAP_DENYWRITE, 3, 0x12000) = 0x7ebc4fe37000
mmap(0x7ebc4fed7000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENY
WRITE, 3, 0xb2000) = 0x7ebc4fed7000
mmap(0x7ebc4ff49000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
|MAP_DENYWRITE, 3, 0x123000) = 0x7ebc4ff49000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0
\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0\0\0"..., 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7e
bc4fc00000
mmap(0x7ebc4fc28000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIX
ED|MAP_DENYWRITE, 3, 0x28000) = 0x7ebc4fc28000
mmap(0x7ebc4fdc0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENY
WRITE, 3, 0x1c0000) = 0x7ebc4fdc0000
mmap(0x7ebc4fe0e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXE
D|MAP_DENYWRITE, 3, 0x20d000) = 0x7ebc4fe0e000
mmap(0x7ebc4fe14000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXE
D|MAP_ANONYMOUS, -1, 0) = 0x7ebc4fe14000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY
|O_CLOEXEC) = 3

```

```

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 70112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7ebc4fb54000
mmap(0x7ebc4fb57000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x7ebc4fb57000
mmap(0x7ebc4fbd3000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x7ebc4fbd3000
mmap(0x7ebc4fbfe000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x7ebc4fbfe000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7ebc4fe23000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7ebc4fb51000
arch_prctl(ARCH_SET_FS, 0x7ebc4fb51b00) = 0
set_tid_address(0x7ebc4fb52128) = 2422
set_robust_list(0x7ebc4fb51de0, 24) = 0
rseq(0x7ebc4fb516a0, 0x20, 0, 0x53053053) = 0
mprotect(0x7ebc4fe0e000, 16384, PROT_READ) = 0
mprotect(0x7ebc4fbfe000, 4096, PROT_READ) = 0
mprotect(0x7ebc4ff49000, 4096, PROT_READ) = 0
mprotect(0x7ebc4ff77000, 4096, PROT_READ) = 0
mprotect(0x7ebc4ffa8000, 4096, PROT_READ) = 0
mprotect(0x5b30b25f7000, 1462272, PROT_READ) = 0
mprotect(0x7ebc4fff6000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("\x7f\xa7\x97\xe8\x49\xfe\x5b\x1f", 8, GRND_NONBLOCK) = 8
munmap(0x7ebc4ffac000, 20151) = 0
statfs("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x404d5005, 0x746af865]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NOEXEC|ST_NOATIME}) = 0

```

```

brk(NULL) = 0x5b30ee7a8000
brk(0x5b30ee7c9000) = 0x5b30ee7c9000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nODEV\tsysfs\nnodev\ttmpfs\nnodev\tbdf"..., 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
read(3, "e 0 0\ndevpts /dev/pts devpts rw,"..., 1024) = 1024
read(3, "relatime 0 0\nmqqueue /dev/mqueue "..., 1024) = 1024
read(3, "s /mnt/wslg/run/user/1000 tmpfs "..., 1024) = 118
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or d
irectory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=
0}, 8) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0
(Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA
_RESTORER|SA_RESTART, sa_restorer=0x7ebc4fc45cb0}, {sa_handler=SIG_DFL
, sa_mask=[], sa_flags=0}, 8) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_
INFINITY}) = 0
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "5b30b1c8e000-5b30b1d80000 r--p 0"..., 1024) = 1024
read(3, "r/lib/x86_64-linux-gnu/libpcr2-"..., 1024) = 1024
read(3, "u/libm.so.6\n7ebc4fe37000-7ebc4fe"..., 1024) = 1024
read(3, "0-7ebc4ff80000 r--p 00000000 08:"..., 1024) = 1024
read(3, "so.2\n7ebc4ffbc000-7ebc4ffeb000 r"..., 1024) = 615

```

```

sched_getaffinity(2422, 32, [0 1 2 3 4 5 6 7 8 9 10 11]) = 8
rt_sigaction(SIGSEGV, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=
0}, 8) = 0
sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_
STACK, -1, 0) = 0x7ebc4ffae000
mprotect(0x7ebc4ffae000, 4096, PROT_NONE) = 0
sigaltstack({ss_sp=0x7ebc4ffaf000, ss_flags=0, ss_size=8192}, NULL) =
0
gettid()
= 2422
rt_sigaction(SIGSEGV, {sa_handler=0x5b30b1fb30c0, sa_mask=[], sa_flags
=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ebc4fc45cb0}, NULL,
8) = 0
rt_sigaction(SIGBUS, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0
}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=0x5b30b1fb30c0, sa_mask=[], sa_flags=
SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ebc4fc45cb0}, NULL,
8) = 0
prctl(PR_GET_AUXV, 0x7ffc2763c270, 0x200, 0, 0) = 448
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_S
TAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, st
x_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/uucore", AT_STATX_SYNC_A
S_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0,
stx_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RD
ONLY|O_CLOEXEC) = 3
statx(3, "", AT_STATX_SYNC_AS_STAT|AT_EMPTY_PATH, STATX_ALL, {stx_mask
=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_
size=4080, ...}) = 0
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32)
= 0
close(3)
= 0
statx(AT_FDCWD, "/usr/share/coreutils/locales/cat", AT_STATX_SYNC_AS_S
TAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, st
x_mode=S_IFDIR|0755, stx_size=4096, ...}) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONL

```

```

rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7ebc4fc45cb0}, {sa_handler=SIG_IGN, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x7ebc4fc45cb0}, 8) = 0
rt_sigaction(SIGSEGV, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7ebc4fc45cb0}, {sa_handler=0x5b30b1fb30c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ebc4fc45cb0}, 8) = 0
rt_sigaction(SIGBUS, {sa_handler=SIG_DFL, sa_mask=~[RTMIN RT_1], sa_flags=SA_RESTORER, sa_restorer=0x7ebc4fc45cb0}, {sa_handler=0x5b30b1fb30c0, sa_mask=[], sa_flags=SA_RESTORER|SA_ONSTACK|SA_SIGINFO, sa_restorer=0x7ebc4fc45cb0}, 8) = 0
ioctl(1, TCGETS2, {c_iflag=ICRNL|IXON, c_oflag=NL0|CR0|TAB0|BS0|VT0|FF0|OPOST|ONLCR, c_cflag=B38400|CS8|CREAD, c_lflag=ISIG|ICANON|ECHO|ECHOE|ECHOK|IEXTEN|ECHOCTL|ECHOKE, ...}) = 0
brk(0x5b30ee7ea000) = 0x5b30ee7ea000
statx(AT_FDCWD, "input.txt", AT_STATX_SYNC_AS_STAT, STATX_ALL, {stx_mask=STATX_ALL|STATX_MNT_ID, stx_attributes=0, stx_mode=S_IFREG|0644, stx_size=22, ...}) = 0
openat(AT_FDCWD, "input.txt", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=22, ...}) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x2), ...}) = 0
splice(3, NULL, 1, NULL, 1048576, 0) = -1 EINVAL (Invalid argument)
pipe([4, 5]) = 0
fcntl(4, F_SETPIPE_SZ, 1048576) = 1048576
splice(3, NULL, 5, NULL, 1048576, 0) = 22
splice(4, NULL, 1, NULL, 22, 0This is a sample file) = 22
splice(3, NULL, 5, NULL, 1048576, 0) = 0
close(5) = 0
close(4) = 0
close(3) = 0
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7ebc4ffae000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
sawdika@DESKTOP-8COA01L:~$ strace cat input.txt 2> trace.txt

```

This is a sample file

```
sadwika@DESKTOP-8CQA01L:~$ cat trace.txt
execve("/usr/bin/cat", ["cat", "input.txt"], 0x7fffc3f94bb8 /* 25 vars
*/) = 0
brk(NULL) = 0x5eac523ab000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1,
0) = 0x7031b950f000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or d
irectory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20151, ...}) = 0
mmap(NULL, 20151, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7031b950a000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY
|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\
0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x703
1b94d7000
mmap(0x7031b94de000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXE
D|MAP_DENYWRITE, 3, 0x7000) = 0x7031b94de000
mmap(0x7031b94ff000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYW
RITE, 3, 0x28000) = 0x7031b94ff000
mmap(0x7031b9506000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
|MAP_DENYWRITE, 3, 0x2f000) = 0x7031b9506000
mmap(0x7031b9508000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED
|MAP_ANONYMOUS, -1, 0) = 0x7031b9508000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O
_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\
0"... , 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x703
1b94a9000
mmap(0x7031b94ad000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXE
```

```

sadwika@DESKTOP-8CQA01L:~$ strace -e trace=openat,read,write,close cat
input.txt
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY
|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O
_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\250\2\0\0\0\0\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY
|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
close(3) = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"..., 1024) = 442
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "none /usr/lib/modules/6.18.33.2-"..., 1024) = 1024
read(3, "e 0 \0\ndevpts /dev/pts devpts rw,"..., 1024) = 1024
read(3, "relatime 0 0\nmqueue /dev/mqueue "..., 1024) = 1024
read(3, "s /mnt/wslg/run/user/1000 tmpfs "..., 1024) = 118

```

```

read(3, "", 1024) = 0
close(3) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "61393e419000-61393e50b000 r--p 0"... , 1024) = 1024
read(3, "0 08:30 13647" "... , 1024) = 1024
read(3, "00 08:30 13650" "... , 1024) = 1024
read(3, " /usr/lib/x86_64-lin"... , 1024) = 1024
read(3, "0 13644 /us"... , 1024) = 566
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "input.txt", O_RDONLY|O_CLOEXEC) = 3
This is a sample file
close(5) = 0
close(4) = 0
close(3) = 0
+++ exited with 0 +++

```